1.a.i

$$J(\theta) = \frac{1}{2}\beta\theta^2 \; ; \; \theta \in R, \alpha \text{ is +ve constant}$$

$$\theta^{[t]} = \theta^{[t-1]} - \alpha \nabla J(\theta^{[t-1]})$$

Let's first find derivative of $J(\theta)$ w.r.t θ

$$\nabla J(\theta) = \frac{d}{d\theta}\left[\frac{1}{2}\beta\theta^2\right] = \beta\theta$$

Clearly, $\nabla J(\theta) = \beta\theta$ is an unbounded in the domain of R

Hence it can't have a Lipschitz Constant.

Let plug in $\nabla J(\theta)$ in the iterative equation

$$\theta^{[t]} = \theta^{[t-1]} - \alpha\beta\theta^{[t-1]}$$

$$\theta^{[t]} = \theta^{[t-1]}\left[1-\alpha\beta\right]$$

As we pass through multiple iterations of this equation $[t=0,1,2\cdots\cdots]$, we expect θ to be approaching to zero which is possible only if $|1-\alpha\beta| < 1$

ie $|1-\alpha\beta| < 1$

$$-1 < 1-\alpha\beta < 1$$

which means

(1) $1-\alpha\beta > -1 \Rightarrow 2 > \alpha\beta \Rightarrow \alpha < 2/\beta$

(2) $1-\alpha\beta < 1 \Rightarrow -\alpha\beta < 0 \Rightarrow \alpha\beta > 0$

(2) which is possible only if α is greater than 0

therefore range of α is as follows:

$$\boxed{0 < \alpha < 2/\beta}$$

$$J(\theta) = \frac{1}{2}\beta\theta^2$$

$$\nabla J(\theta) = \frac{d}{d\theta}\left[\frac{1}{2}\beta\theta^2\right] = \beta\theta.$$

$\frac{1}{2}\beta\theta^2$ is a convex function at minimum,

$$\nabla J(\theta) = 0$$

$$\therefore \beta\theta = 0 \Rightarrow \theta = 0.$$

i.e $J(\theta)$ converges at $\underline{\theta = 0}$

ie $\underline{\theta^{\dagger} = 0}$

1. a. ii

$$\theta^{[t]} = \theta^{[t-1]} - \alpha\beta\theta^{[t-1]}$$

T is the minimum number of iterations that we have to compute for convergence such that $|\theta^{[T]} - \theta^* \le \varepsilon|$

$$\theta^{[T]} = \theta^{[0]}(1-\alpha\beta)^T$$

Applying Logarithm on both sides

$$\log \theta^{[T]} = \log \theta^{[0]} + T \log(1-\alpha\beta)$$

$$T = \frac{\log \theta^{[T]} - \log \theta^{[0]}}{\log(1-\alpha\beta)}$$

we know α' range is $0 < \alpha < 2/\beta$

1. with in $0 < \alpha < 1/\beta$, as α increases, $1-\alpha\beta$ gets more closer to zero. Hence the value of denominator $\log(1-\alpha\beta)$ is negative and decreases. Hence the value of T increases with α increasing within the range of $0 < \alpha < 1/\beta$.

2. Similarly, with in the range $1/\beta < \alpha < 2/\beta$ the value of T decreases as α increases

simply put $(0, 1/\beta) \rightarrow \alpha$ increases, T increases
$(1/\beta, 2/\beta) \rightarrow \alpha$ increases, T decreases

1 · a · ii

It's very important to chose the appropriate α for the proper convergence at the minimum values as closely as possible such that $|\theta^{[T]} - \theta^{*}| \leq \epsilon$.

* Larger values of α makes the computation to completely ignore (or) step over the expected minimal value range (ie. $|\theta^{[T]} - \theta^{*}| \leq \epsilon$ thereby not serving the need of computing acceptable minimum value.

* smaller values of α makes it very time consuming and compute intensive to reach the convergence such that $|\theta^{[T]} - \theta^{*}| \leq \epsilon$.

Hence its important to chose the right value for α.